

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

InventralQ

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**InventralQ**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Swati Goel. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place : Rajpura

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Date : 10/05/2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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We are also thankful to Chitkara University for providing us with the necessary infrastructure, technical resources, and learning environment required for the successful execution of the project. The knowledge and experience gained during our academic journey greatly contributed to the development of InventraIQ.

We would also like to acknowledge the efforts and cooperation of all team members whose dedication and teamwork made this project possible. Finally, we express our heartfelt gratitude to our parents and family members for their continuous motivation, encouragement, and moral support throughout the project.

1. Introduction

1.1 Problem Background

Inventory management is an essential part of any business organization, especially for small and medium-sized enterprises that handle large numbers of products and daily transactions. Many businesses still rely on traditional methods such as manual record keeping, paper-based documentation, or spreadsheets like Microsoft Excel to manage their inventory. Although these methods are simple to use, they often become inefficient as the business grows and the amount of data increases. Manual systems are more likely to produce errors such as incorrect stock records, duplicate entries, delayed updates, and data inconsistency.

Another major challenge faced by businesses is the lack of real-time inventory tracking. In traditional systems, inventory updates are not reflected instantly, making it difficult for multiple users to access the latest stock information simultaneously. This can lead to problems such as overstocking, stock shortages, and poor inventory planning, which directly affect business operations and profitability. Additionally, most conventional systems do not provide intelligent analysis or future demand forecasting, making it difficult for businesses to make informed decisions regarding restocking and inventory control.

To overcome these limitations, there is a need for a smart and real-time inventory management solution that can automate inventory operations, improve data accuracy, and support multiple users efficiently. Integrating machine learning techniques into the system can further enhance decision-making by analyzing historical sales data and predicting future product demand. This helps businesses optimize inventory levels and improve overall operational efficiency.

1.2 Objective of the Project

InventraIQ is an organization's web-based solution for managing and monitoring stock levels. The main objective of InventraIQ is to provide a consolidated inventory management system that facilitates accurate and timely tracking of stock levels, with the least amount of chance for human error associated with manual entry. This new platform will update stock levels automatically using automated processes when specific events occur (i.e., sales, purchases, returns and transfers).

To accommodate multiple users, the InventraIQ application has a dedicated user interface, which including login capabilities for each person with a unique user account and access controls set according to an employee's role. This allows for individualized security through the entire application. The application is also designed to be flexible enough to meet any potential organization's future growth requirements through the ability to adapt easily with minimal disruption to existing users.

Lastly, integrating predictive analytics through machine learning ('ml') technology will provide InventraIQ with the opportunity to analyze historical sales (data) and realize better sales forecasts based on today's and past demand patterns. By using predictive analytics, InventraIQ will provide organizations the opportunity to minimize their chances of running out of products they are trying to sell (understocking) or carrying excess inventory (overstocking). The project will consist of bringing machine learning (ml) methodology into InventraIQ, with the intent of producing a prediction model that can produce forecasts for the next 30-days based on historical sales data.

1.3 Methodology

Frontend Design:

For front-end development of InventraIQ; ReactJS and Tailwind CSS were used to make a user-friendly and responsive experience that provides fast, smooth navigation and interactions and enables users to manage their products, sales and inventory data quickly and easily. Flexible layouts enable users to navigate on various types of devices and display sizes with effective consistency throughout their experience by having re-usable UI components.

Backend Design:

NodeJS and Express JS have been used to implement functionality necessary for managing all server-side operations and providing an efficient means of communication between the servers and frontend via the use of APIs. The Backend Services (authentication, product management, inventory updates, sales processing); are all handled by the backend; and the (frontend, database, and Machine Learning) communicate with each other via the use of RESTful APIs, providing a way for the front end and backend to provide a consistently secure and scalable performance for the application.

Database Design:

MongoDB is being used as the main database for storing (managing) Users' details, Products, Inventory and Sale Transactions. Mongoose ORM has been integrated with the Database Layer in order to simplify interaction with the database and provide better management of the database schemas. The manner in which the database is structured allows for efficient retrieval of data, provides real-time data updates and supports the ongoing scalability required to support the growth of inventory data being contained within the database.

ML Service:

The provision of demand prediction and inventory analysis is being performed by a Flask-based microservice that is integrated within the overall system. Historical sales data are processed (using Python Libraries such as Pandas) against the predicted demand using Machine Learning Algorithms.

1.4 Tools and Technologies

Frontend: React.js (User Interface Development)

Backend: Node.js with Express.js (Server-side Development)

Database: MongoDB (Data Storage and Management)

Machine Learning Service: Flask (Python-based ML Microservice)

Machine Learning Library: Scikit-learn (Linear Regression Model)

Data Processing: Pandas and NumPy

Authentication: JWT (JSON Web Token)

Version Control: Git and GitHub

Code Editor: Visual Studio Code

Runtime Environment: Node.js Runtime Environment

1.5 Key Features or Functionality

This system has multiple capabilities to enhance the management of inventory:

- Secure log-in using JWT authentication
- Role-based access control: Admin, Manager, Staff
- Manage products: add, modify, view, delete
- Real-time stock levels
- Low-stock alerts
- reorder quickly - Dead stock detection
- 30 days demand forecasting

1.6 Expected Outcome

The system should provide:

- Centralised inventory control
 - Real time inventory updates
 - Automatic transaction recording
 - Better accuracy and less manual errors
 - More informed decisions made possible by using data-driven insights
- Also, a machine- learning capability will assist in planning inventory for better management and stock issues. The machine learning feature further helps in planning inventory effectively and avoiding stock-related issues.

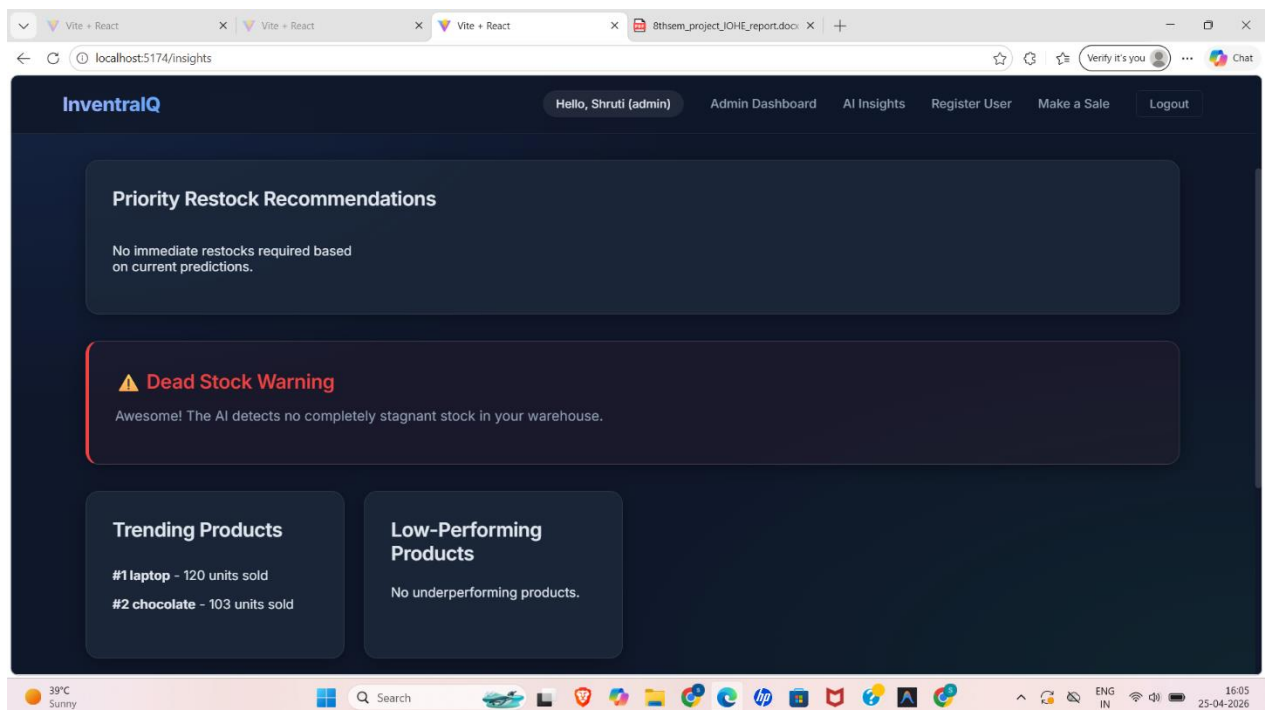
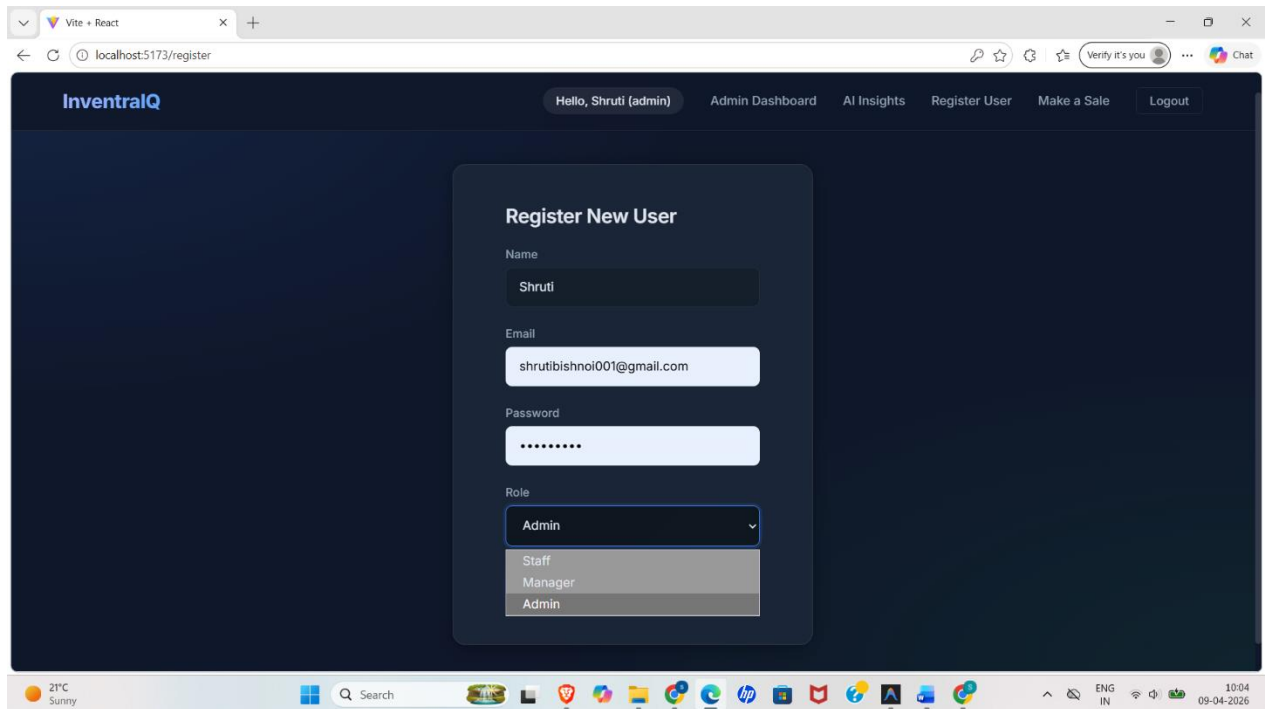
2. Conclusion

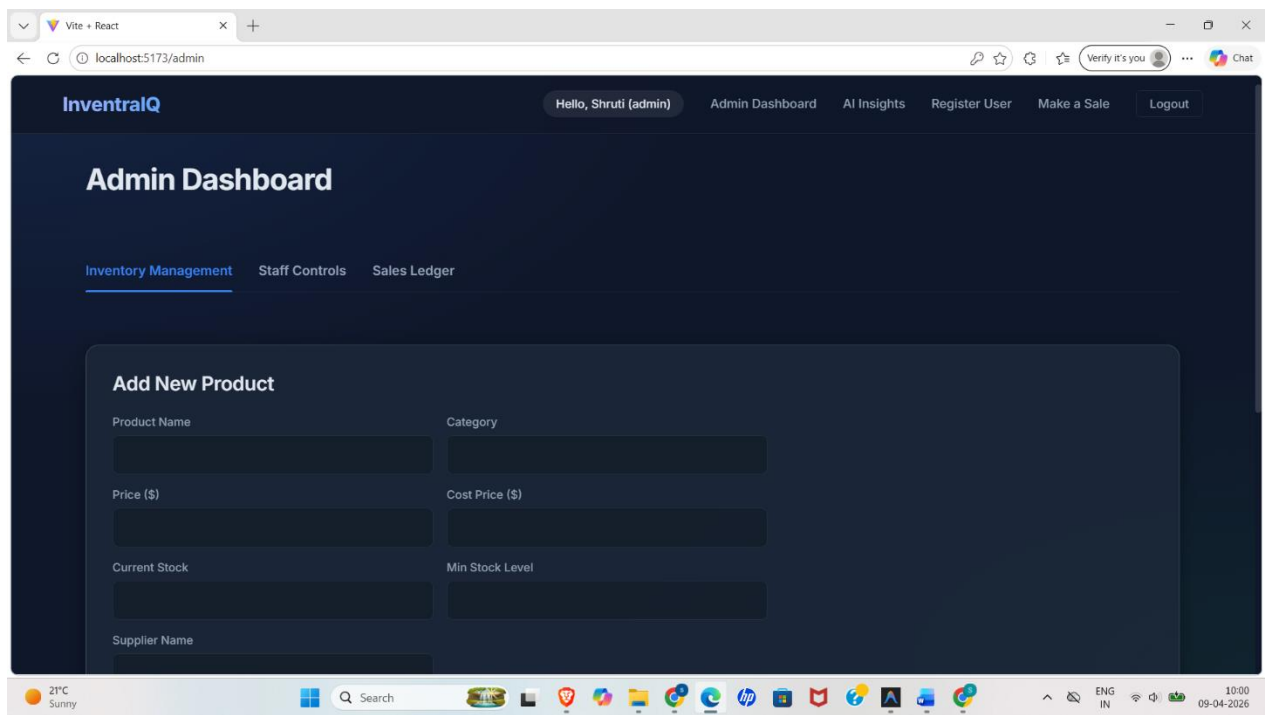
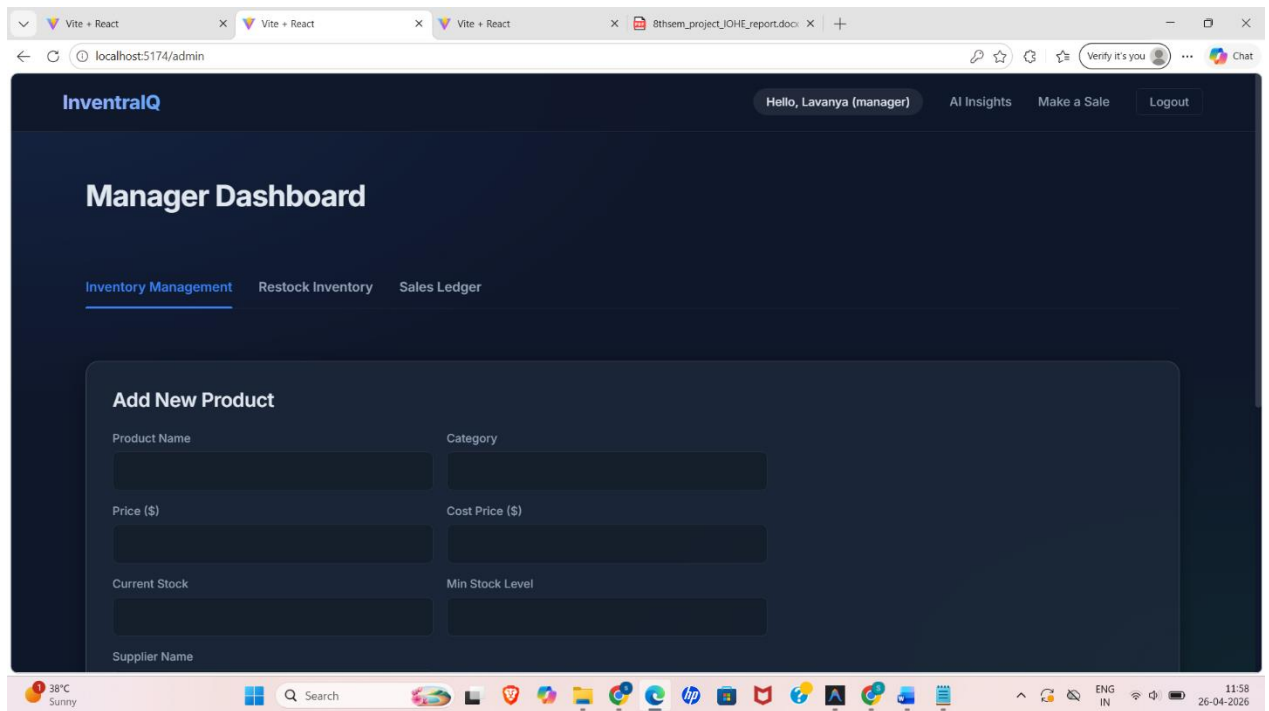
InventraIQ is a smart and user-friendly inventory management system designed to improve the efficiency and accuracy of inventory operations for businesses. The system provides real-time inventory updates, product management, and sales tracking, helping organizations reduce manual work, minimize human errors, and maintain accurate stock records. By centralizing inventory-related activities into a single platform, the system makes it easier for users to monitor stock movement, manage products, and track sales transactions efficiently.

The project is developed using modern technologies such as React.js, Node.js, Express.js, MongoDB, and Flask, which together provide a robust, scalable, and responsive platform for inventory management. The frontend offers an intuitive and interactive user experience, while the backend efficiently handles server-side operations, authentication, and API communication. MongoDB ensures secure and efficient storage of product, user, and sales data, allowing smooth data management and retrieval.

A key feature of InventraIQ is the integration of machine learning for demand prediction and inventory analysis. The Flask-based machine learning service analyzes historical sales data using linear regression techniques to predict product demand for the next 30 days. This predictive capability helps businesses make informed decisions regarding inventory planning, restocking, and product management. By combining real-time inventory tracking with predictive analytics, InventraIQ provides an effective, intelligent, and reliable solution for modern inventory management systems.

3. Output





Vite + React

localhost:5173/insights

Verify it's you

Chat

InventraIQ

Hello, Shruti (admin)

Admin Dashboard

AI Insights

Register User

Make a Sale

Logout

Trending Products

#1 laptop - 90 units sold

#2 chocolate - 42 units sold

Low-Performing Products

No underperforming products.

All Product Predictions

PRODUCT	STOCK	30-DAY DEMAND	STATUS	AI MESSAGE
laptop	0	0	Restock Needed	Critical: Stock has fallen below the mandatory minimum threshold
chocolate	18	1260	Restock Needed	ML flags high risk of stockout based on predicted demand trend

21°C Sunny

Search

hp

ENG IN

10:00 09-04-2026

4. References

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- Node.js – JavaScript runtime environment.
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- Available at: <https://flask.palletsprojects.com>